Convex Optimization - Exercises from Chapter 3 of Boyd

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3.1

(a)$f : \mathbb{R} \to \mathbb{R}$ convex, $a, b \in \mathbf{dom}\ f$ with $a < b$

Show that: (a) $f(x) \leq \left(\frac{b-x}{b-a}\right) f(a) + \left(\frac{x-a}{b-a}\right) f(b)$

f convex.
By Jensen's inequality : $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y)$
For any x in $[a, b]$

$$C_1 + C_2 = \frac{b-x}{b-a} + \frac{x-a}{b-a} = 1$$

Applying Jensen's inequality:
$f(C_1 \cdot a + C_2 \cdot b) \leq C_1 \cdot f(a) + C_2 \cdot f(b)$

$$\begin{aligned} C_1 \cdot a + C_2 \cdot b &= \frac{b-x}{b-a} \cdot a + \frac{x-a}{b-a} \cdot b \\ &= \frac{ab - ax + bx - ab}{b-a} \\ &= \frac{b-a}{b-a} \cdot x = x \end{aligned}$$

$$f(x) = f(C_1 \cdot a + C_2 \cdot b) \leq C_1 \cdot f(a) + C_2 \cdot f(b)$$

(b) $\frac{f(x)-f(a)}{x-a} \leq \frac{f(b)-f(a)}{b-a} \implies$

$f(x) - f(a) \leq \frac{x-a}{b-a} f(b) + \left(1 - \frac{x-a}{b-a}\right) f(a) \implies$

$f(x) \leq \underbrace{\left(\frac{x-a}{b-a}\right)}_{C_1} \cdot f(b) + \underbrace{\left(\frac{b-x}{b-a}\right)}_{C_2} f(a)$

$C_1 + C_2 = \frac{x-a+b-x}{b-a} = 1$

Correct because $f(x)$ is convex

$$\frac{f(b)-f(a)}{b-a} \leq \frac{f(b)-f(x)}{b-x} \implies -\frac{f(b)-f(x)}{b-x} \leq -\frac{f(b)-f(a)}{b-a} \implies$$

$$\implies f(x) - f(b) \leq -\frac{b-x}{b-a} \cdot (f(b) - f(a)) \implies$$

$$\implies f(x) \leq f(b) \cdot \left(1 - \frac{b-x}{b-a}\right) + f(a) \cdot \frac{b-x}{b-a} \implies$$

$$\implies f(x) \leq f(b) \cdot \underbrace{\left(\frac{x-a}{b-a}\right)}_{C_1} + f(a) \cdot \underbrace{\left(\frac{b-x}{b-a}\right)}_{C_2} \implies$$

With $C_1 + C_2 = 1$. Thus proved.

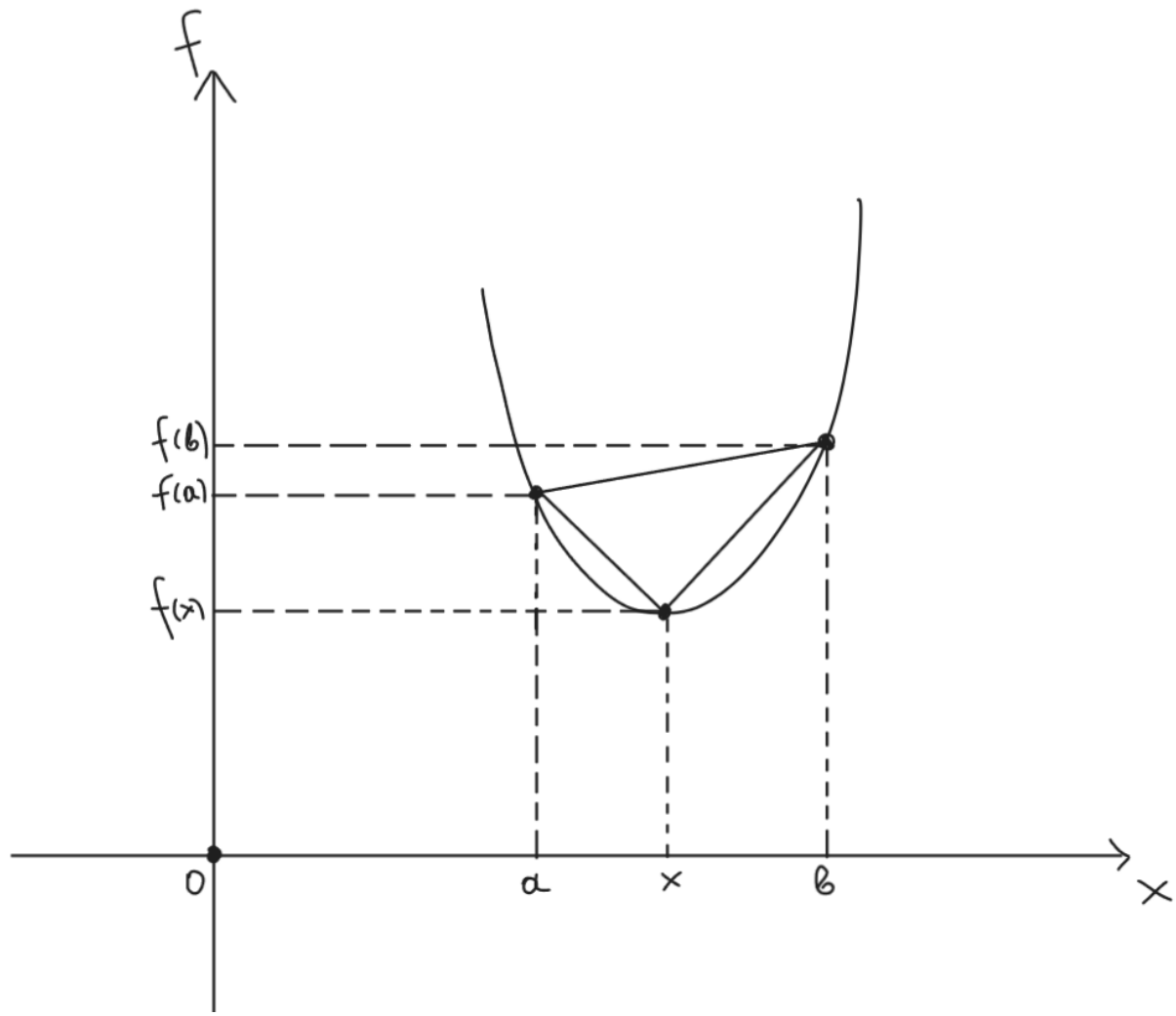

Figure 1: Inequality of the slopes of intermediate point

(c) For the left hand of the inequality: $\forall x \in (a,b)$

$\frac{f(x)-f(a)}{x-a} \leq \frac{f(b)-f(a)}{b-a} \implies$

$\implies \frac{f(a+h)-f(a)}{a+h-a} \leq \frac{f(b)-f(a)}{b-a} \implies$

$\implies \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h} \leq \frac{f(b)-f(a)}{b-a}$

If f differentiable, the limit exists and is equal to $f'(a) \implies f'(a) \leq \frac{f(b)-f(a)}{b-a}$

For the right hand of the inequality: $\forall x \in (a,b)$

$\frac{f(b)-f(a)}{b-a} \leq \frac{f(b)-f(x)}{b-x} \implies$

$\implies \frac{f(b)-f(a)}{b-a} \leq \frac{f(b)-f(b-h)}{b-(b-h)} = \frac{f(b)-f(b-h)}{h} \implies$
replace $h \to -k$

$\implies \frac{f(b)-f(a)}{b-a} \leq \frac{f(b+k)-f(b)}{k} \implies$

$\implies \frac{f(b)-f(a)}{b-a} \leq \lim_{k\to 0^-} \frac{f(b+k)-f(b)}{k} \implies$

$\implies \frac{f(b)-f(a)}{b-a} \leq f'(b)$

(d) Suppose that f is twice differentiable. Without loss of generality, we can write the previous inequality for any a, b.

For fixed a, and $h > 0$ for $(a, a+h) \in \text{dom } f$

$f'(a) \leq \frac{f(b)-f(a)}{b-a} \leq f'(b)$
From monotonicity:
$f'(a) \leq \frac{f(a+h)-f(a)}{h} \leq f'(a+h) \implies$

$\implies f'(a+h) - f'(a) \geq 0 \implies \frac{f'(a+h)-f'(a)}{h} \geq 0 \implies \lim_{h\to 0} \frac{f'(a+h)-f'(a)}{h} \geq 0$

$\implies f''(a) \geq 0$

For fixed b, and $h > 0$, for $(b-h, b) \in \text{dom } f$:
$\frac{f(b)-f(a)}{b-a} \leq \frac{f(b)-f(b-h)}{h} \leq f'(b)$.
Replacing $h = -k$:
$\implies \frac{f(b+k)-f(b)}{k} \leq f'(b) \implies \lim_{k\to 0^-} \frac{f(b+k)-f(b)}{k} \leq f'(b)$.
Since $b + k < b$ for $k < 0$, the monotonicity of the gradient implies $f'(b+k) \leq f'(b)$.
$\implies f'(b) - f'(b+k) \geq 0 \implies \frac{f'(b+k)-f'(b)}{k} \geq 0 \quad$ (since $k < 0$).
$\implies f''(b) = \lim_{k\to 0^-} \frac{f'(b+k)-f'(b)}{k} \geq 0$.

3.3

For $x_1 < x_2 \implies f(x_1) < f(x_2)$
For $y_1 < y_2 \implies g(y_1) < g(y_2)$
g is increasing $\circledast$

Let $x_1 = g(y_1)$, $x_2 = g(y_2)$
$f(x_1) = y_1$, $f(x_2) = y_2$

Combine: $x_\theta = \theta x_1 + (1-\theta)x_2, \quad \theta \in [0,1]$

f convex $\implies f(x_\theta) \leq \theta f(x_1) + (1-\theta) f(x_2) \implies$

$\overset{\circledast}{\implies} g(f(x_\theta)) \leq g(\theta y_1 + (1-\theta) y_2) \implies$

$\implies \theta x_1 + (1-\theta)x_2 = \theta \cdot g(y_1) + (1-\theta) g(y_2) \leq g(\theta y_1 + (1-\theta) y_2)$

Thus g is concave.

3.6

(a) Let $y \in \mathbb{R}^{n+1} \implies y = (x, x_{n+1})$ with $x \in \mathbb{R}^n, x_{n+1} \in \mathbb{R}$

Halfspace: for $\alpha \neq 0$

$$H = \left\{(x, x_{n+1}) \mid \alpha^T y \leq b, \alpha \in \mathbb{R}^{n+1}, b \in \mathbb{R}\right\}$$

for $\alpha^T = [\alpha' \quad \alpha_{n+1}]$ with $\alpha' \in \mathbb{R}^n, \alpha_{n+1} \in \mathbb{R}$

$H \equiv \{(x, x_{n+1}) \mid \alpha' x + \alpha_{n+1} x_{n+1} \leq b\}$

$H \equiv \{(x, x_{n+1}) \mid \alpha_{n+1} x_{n+1} \leq b - \alpha' x\}$

For $a \neq 0$:

For positive $\alpha_{n+1} \to H \equiv \left\{(x, x_{n+1}) \mid x_{n+1} \leq \frac{b - \alpha' x}{\alpha_{n+1}}\right\}$

For negative $\alpha_{n+1} \to H \equiv \left\{(x, x_{n+1}) \mid x_{n+1} \geq \frac{b - \alpha' x}{\alpha_{n+1}}\right\}$

In both cases H is a halfspace, and we can define $f(x) = \frac{b}{\alpha_{n+1}} - \frac{\alpha'}{\alpha_{n+1}} x$ which is an affine function.

For the set to be equivalent to an epigraph of a function, we must choose only strictly negative a_{n+1}.

In conclusion for an epigraph of a function to be a halfspace, the function must be affine.

(b) $epif =$ cone

$epif = \{(x, t) \mid x \in \mathbb{R}^n, t \geq f(x)\}$

Cone: $\forall\ x \in C \implies \theta \cdot x \in C$ for $\theta \geq 0$

$\theta \cdot (x, t) = (\theta x, \theta t) \in C$

For $epif =$ cone:

for $t \geq f(x) \implies \theta t \geq f(\theta x)$

On the boundary $t = f(x) \implies \theta \cdot f(x) \geq f(\theta x) \quad (1)$

For $x' = \theta x$, $\theta \cdot f(x) \geq f(x')$.

And from the cone property at the boundary of $epif$:

$$\frac{1}{\theta} \cdot f(x') \geq f\left(\frac{1}{\theta} x'\right) \implies$$
$$\frac{1}{\theta} f(\theta x) \geq f(x) \implies$$
$$f(\theta x) \geq \theta f(x) \quad (2)$$

From (1), (2):

$$\left.\begin{array}{l} \theta f(x) \geq f(\theta x) \\ f(\theta x) \geq \theta f(x) \end{array}\right\} \implies \theta f(x) = f(\theta x) \quad \circledast$$

For $epif$ to be a convex cone, the convex property must be satisfied:

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2) \quad \circledast\circledast$$

for $x_1, x_2 \in C$.

Thus the $epif$ is a convex cone iff:

$$\theta f(x) = f(\theta x)$$
$$\text{and}$$
$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2)$$

(c) $epif =$ polyhedron

A polyhedron is an intersection of a finite number m of constraints and inequalities.

$$P = \bigcap_{i=1}^{m} \left\{ {a^i}^T x \leq b,\ Cx = d, x \in \mathbb{R}^{n+1} \right\}$$

$epif = \{(x,t) \mid x \in \text{dom } f \subseteq \mathbb{R}^n,\ t \geq f(x)\}$

Let $a^i = \begin{bmatrix} a \\ a_{n+1} \end{bmatrix}$

$a \cdot x + a_{n+1} \cdot x_{n+1} \leq b \implies$

$a_{n+1} \cdot x_{n+1} \leq b - ax$

- $x_{n+1} \leq \frac{b-ax}{a_{n+1}}, \quad a_{n+1} > 0$

- $x_{n+1} \geq \frac{b-ax}{a_{n+1}}, \quad a_{n+1} < 0$

In this case with $a_{n+1} = 0$, the domain of f does not include x_{n+1} .

We discard the first inequality, because the epigraph is unbounded from above.

We write the i-th inequality as:

$$x_{n+1} \geq \frac{b-ax}{a_{n+1}} = \frac{b}{a_{n+1}} - \frac{a}{a_{n+1}}x$$

which is an affine function

$$f_i(x) = \frac{b}{a_{n+1}} - \frac{a}{a_{n+1}}x$$

$x_{n+1} \geq f_i(x)$
$t \geq f_i(x)$

For $i = 1, \ldots, m$ we get m inequalities that must hold:

$$t \geq f_1(x), \quad t \geq f_2(x), \ldots, \quad t \geq f_m(x)$$

For all inequalities to hold simultaneously,:

$$t \geq \max\{f_1(x), \ldots, f_m(x)\}$$

3.7

Suppose $f \neq c$
$\exists\, x, y : f(x) < f(y) \iff f(y) - f(x) > 0$

Let $z(t) = (1-t)x + ty$, for $t > 1$
$z \in \text{dom } f$, because dom $f = \mathbb{R}^n$

We can write:

$$y = \left(1 - \frac{1}{t}\right) x + \frac{1}{t} z(t)$$

By convexity of f:

$$f(y) \leq \left(1 - \frac{1}{t}\right) f(x) + \frac{1}{t} f(z(t)) \implies$$

$$\implies f(z(t)) \geq t \cdot f(y) - (t-1) f(x) = f(x) + t \underbrace{(f(y) - f(x))}_{>0}$$

We can arbitrarily choose t, which gives:

For $t \to \infty \implies f(z(t)) \to \infty$.

This contradicts f being bounded above.

Thus $f = c$.

3.16

(a) $f(x) = e^x - 1$ on $\mathbb{R}$

$h(x) = e^x$ is strictly convex.
$g(x) = -1$ is affine.
$f(x) = h(x) + g(x)$ is convex as a non-negative sum of convex functions.

$f(x)$ is quasiconvex because it is convex.

Since $f(x) = e^x$ is strictly monotonic, it is by definition both quasiconcave and quasiconvex.

(b) $f(x_1, x_2) = x_1 \cdot x_2$ on $\mathbb{R}^2_{++}$

$\partial_{x_1} f = x_2$
$\partial_{x_2} f = x_1$
$\partial_{x_1 x_1} f = 0$
$\partial_{x_2 x_2} f = 0$
$\partial_{x_1 x_2} f = 1$
$\partial_{x_2 x_1} f = 1$

$$\nabla^2 f = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \mathcal{H}$$

For Hessian to be semidefinite:
$x^T \mathcal{H} x \geq 0$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y & x \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2xy$$

For positive semidefiniteness, the inequality $2xy \geq 0$ must hold for any vector. Which it does not, i.e., for $x = 1, y = -1$.

Thus f is not convex.
With similar steps we can show it is not concave either.

An unnecessarily long **proof at the end of the file.**

Conclusively, f is not quasiconvex. With exactly symmetric reasoning, we conclude that f is quasiconcave, by showing that -f , has convex sublevel sets.

(c) $f(x, y) = \frac{1}{xy}$

$$\nabla^2 f(x, y) = H = \begin{bmatrix} \frac{2}{x^2y^2} \cdot \frac{y}{x} & \frac{1}{x^2y^2} \\ \frac{1}{x^2y^2} & \frac{2}{x^2y^2} \cdot \frac{x}{y} \end{bmatrix} \implies \det H = \frac{4}{x^4y^4} \cdot \frac{yx}{xy} - \frac{1}{x^4y^4} = \frac{1}{x^4y^4} \cdot 3 > 0$$

Thus f is strictly convex, therefore quasiconvex.
Also, f is not concave nor affine.

$f(x_1, x_2) = \frac{1}{x_1 x_2} \quad , \quad x_1, x_2 \in \mathbb{R}^2_{++}$

$f(x,y) = \frac{1}{xy} \longrightarrow g = -f = -\frac{1}{xy}$

$S_a(g) = \{x, y \mid -\frac{1}{xy} \leq a \ , \ x, y \in \mathbb{R}^2_{++}\}$

$-\frac{1}{xy} \leq a \quad , \quad \text{with } x, y > 0$

$$\frac{1}{xy} \geq -a \implies \begin{cases} \text{For } a > 0 : \frac{1}{-ax} \leq y \text{ . This can be defined by the epigraph of } h_1(x) = -\frac{1}{ax} \text{ , with } a > 0 \\ \\ \text{For } a < 0 : \frac{1}{-ax} \geq y \implies \frac{1}{ax} \leq y \text{ . Let } u = -y \implies \text{epi } h_2(u). \text{ With } h_2 = \frac{1}{ax} = u \text{ , with } a < 0, x > 0. \end{cases}$$

Both subsets must be convex for convexity of S_a.

$h_2(x) = \frac{1}{ax} \quad , \quad a < 0 \ , \ x > 0$

Is this a convex function?
Let $k = -a > 0$
$h_2(x) = -\frac{1}{kx}$
$\nabla^2 h_2 = -\frac{2}{kx^3} < 0$

Thus h_2 is not convex.

This means that f is not quasiconcave.

(d) $f(x,y) = \frac{x}{y}$ on $\mathbb{R}^2_{++}$

$\partial_x f = \frac{1}{y}$
$\partial_{xx} f = 0$
$\partial_{xy} f = -\frac{1}{y^2}$
$\partial_y f = -\frac{x}{y^2}$
$\partial_{yx} f = -\frac{1}{y^2}$
$\partial_{yy} f = \frac{2x}{y^3}$

$$\nabla^2 f = H = \begin{bmatrix} 0 & -\frac{1}{y^2} \\ -\frac{1}{y^2} & \frac{2x}{y^3} \end{bmatrix} \implies \det H = -\frac{1}{y^4}$$

The function is neither convex nor concave.

Sublevel sets f:
$S_\alpha = \{(x,y) \mid f(x,y) \leq \alpha, \ (x,y) \in \text{dom } f\} \equiv \left\{(x,y) \mid \frac{x}{y} \leq \alpha, \ (x,y) \in \text{dom } f\right\}$

We check for convexity.
$S_\alpha \equiv \{(x,y) \mid x \leq \alpha y \implies x - \alpha y \leq 0 \implies A(x,y) \leq 0,\ (x,y) \in \text{dom } f\}$
$A := \begin{bmatrix} 1 & -\alpha \end{bmatrix}$

The set is convex because it is a linear combination. For quasiconcavity: $f' = -f = -\frac{x}{y} \implies S'_\alpha = \left\{(x,y) \mid -\frac{x}{y} \leq \alpha,\ (x,y) \in \text{dom } f\right\} \equiv \{(x,y) \mid Ax \geq 0,\ (x,y) \in \text{dom } f\}$

Thus f is quasiconcave because the sublevel sets of $-f$ are convex.

(e)

$f(x_1, x_2) = \frac{x_1^2}{x_2} \quad , \quad \mathbb{R} \times \mathbb{R}_{++}$

Hessian:

$$\begin{aligned}
\partial_x f &= \frac{2x}{y} \\
\partial_{xx} f &= \frac{2}{y} \\
\partial_{xy} f &= -\frac{2x}{y^2} \\
\partial_y f &= -\frac{x^2}{y^2} \\
\partial_{yx} f &= -\frac{2x}{y^2} \\
\partial_{yy} f &= \frac{2x^2}{y^3}
\end{aligned}
\qquad
H = \begin{bmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & \frac{2x^2}{y^3} \end{bmatrix} = 2\frac{x^2}{y^2} \begin{bmatrix} \frac{y}{x^2} & -\frac{1}{x} \\ -\frac{1}{x} & \frac{1}{y} \end{bmatrix}$$

$\det H = 4\frac{x^4}{y^4} \cdot \left(\frac{1}{x^2} - \frac{1}{x^2}\right) \implies \det H = 0$

Semidefinite

$$x^T H x = \begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & \frac{2x^2}{y^3} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 2 \begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} \frac{u}{y} - \frac{x}{y^2} v \\ -\frac{xu}{y^2} + \frac{x^2 v}{y^3} \end{bmatrix} =$$

$$= 2\left(\frac{u^2}{y} - \frac{x \cdot uv}{y^2} - \frac{x \cdot uv}{y^2} + \frac{x^2 v^2}{y^3}\right) = 2\left(\frac{u^2}{y} - \frac{2uvx}{y^2} + \frac{x^2 v^2}{y^3}\right) =$$

$$= \frac{2}{y^3}\left[u^2 y^2 - 2uvxy + v^2 x^2\right] =$$

$$= \frac{2}{y^3}(uy - vx)^2 \geq 0$$

Thus H is positive semidefinite. We conclude that f is convex and quasiconvex , not concave or affine.

For quasiconcavity :

$f(x_1, x_2) = \frac{x_1^2}{x_2}$

$g = -f = -\frac{x_1^2}{x_2}$

$S_a(g) = \{(x_1, x_2) \mid -\frac{x_1^2}{x_2} \leq a \ , \ (x_1, x_2) \in \text{dom } g\}$

$\equiv \{(x_1, x_2) \mid \frac{x_1^2}{x_2} \geq -a \ , \ (x_1, x_2) \in \text{dom } g\}$

$\equiv \{(x_1, x_2) \mid x_1^2 \geq -ax_2 \ , \ (x_1, x_2) \in \text{dom } g\}$

$$x_1^2 \geq -ax_2 \longrightarrow \begin{cases} \text{For } a = 0 : \frac{x_1^2}{x_2} \geq 0 \implies \mathbb{R} \times \mathbb{R}_{++}, \text{ Convex} \\ \\ \text{For } a > 0 : x_1^2 \geq -ax_2 \implies x_2 \geq -\frac{x_1^2}{a}. \text{ Let } h_1(x_1) = -\frac{x_1^2}{a} \ , \ a > 0, x_1 \in \mathbb{R}. \\ \qquad \text{This holds for every } x_1, x_2 \text{ in } dom\ g. \text{ Thus } epi h_1 \text{ is a convex set and } h_1 \text{ is convex.} \\ \\ \text{For } a < 0 : x_1^2 \geq -ax_2 \implies x_1^2 \geq kx_2 \implies x_2 \leq \frac{x_1^2}{k}, \text{ with } k > 0. \\ \qquad \text{Let } h_2(x_1) = \frac{x_1^2}{k}. \text{ This is the hypograph of a convex function,} \\ \qquad \text{thus non-convex for } a < 0. \end{cases}$$

This means that S_a of g are not always convex.
Thus f is not quasiconcave.

(f)

$f(x, y) = x^\alpha y^{1-\alpha} \quad , \quad \alpha \in [0, 1] \ , \ (x, y) \in \mathbb{R}^2_{++}$

Hessian: $\nabla^2 f$

$$\begin{aligned} \partial_x f &= \alpha x^{\alpha-1} y^{1-\alpha} \\ \partial_{xx} f &= \alpha(\alpha-1) x^{\alpha-2} y^{1-\alpha} \\ \partial_{xy} f &= \alpha(1-\alpha) x^{\alpha-1} y^{-\alpha} \\ \partial_y f &= (1-\alpha) x^\alpha y^{-\alpha} \\ \partial_{yx} f &= (1-\alpha)\alpha x^{\alpha-1} y^{-\alpha} \\ \partial_{yy} f &= (1-\alpha)(-\alpha) x^\alpha y^{-(1+\alpha)} \end{aligned}$$

$$\nabla^2 f = H = \begin{bmatrix} \alpha(\alpha-1)x^{\alpha-2}y^{1-\alpha} & \alpha(1-\alpha)x^{\alpha-1}y^{-\alpha} \\ (1-\alpha)\alpha x^{\alpha-1}y^{-\alpha} & (1-\alpha)(-\alpha)x^\alpha y^{-(1+\alpha)} \end{bmatrix}$$

$$\det H = (\alpha)(-\alpha)(1-\alpha)(\alpha-1) \cdot x^{\alpha} x^{\alpha-2} y^{1-\alpha} y^{-(1+\alpha)} - (\alpha \cdot \alpha \cdot (1-\alpha)(1-\alpha) \cdot x^{\alpha-1} x^{\alpha-1} y^{-\alpha} y^{-\alpha})$$

$$= \alpha^2(\alpha-1)^2 \left[(x^{\alpha-1})^2 (y^{-\alpha})^2 - (x^{\alpha-1})^2 (y^{-\alpha})^2 \right] = 0$$

Semidefinite check : $x^T H x = \begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} \alpha(\alpha-1)x^{\alpha-2}y^{1-\alpha} & \alpha(1-\alpha)x^{\alpha-1}y^{-\alpha} \\ (1-\alpha)\alpha x^{\alpha-1}y^{-\alpha} & (1-\alpha)(-\alpha)x^{\alpha}y^{-(1+\alpha)} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$

$$= \begin{bmatrix} u \cdot \alpha(\alpha-1)x^{\alpha-2}y^{1-\alpha} + v(1-\alpha)\cdot\alpha\cdot x^{\alpha-1}y^{-\alpha} & u\cdot\alpha(1-\alpha)\cdot x^{\alpha-1}y^{-\alpha} + v(1-\alpha)(-\alpha)\cdot x^{\alpha}y^{-(1+\alpha)} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} =$$

$$= u^2\alpha(\alpha-1)x^{\alpha-2}y^{1-\alpha} + uv\alpha(1-\alpha)x^{\alpha-1}y^{-\alpha} + uv\alpha(1-\alpha)x^{\alpha-1}y^{-\alpha} + v^2(1-\alpha)(-\alpha)x^{\alpha}y^{-(1+\alpha)} =$$

$$= -(1-\alpha)\alpha\cdot u^2\cdot x^{\alpha-2}y^{1-\alpha} + 2uv\cdot(1-\alpha)\alpha\cdot x^{\alpha-1}y^{-\alpha} - (1-\alpha)\alpha\cdot v^2\cdot x^{\alpha}y^{-(1+\alpha)} =$$

$$= -(1-\alpha)\alpha\left[u^2\cdot\tfrac{x^\alpha}{y^\alpha}\cdot\tfrac{y}{x^2} - 2uv\cdot\tfrac{x^\alpha}{y^\alpha}\cdot\tfrac{1}{x} + \tfrac{x^\alpha}{y^\alpha}\cdot\tfrac{1}{y}\cdot v^2\right] =$$

$$= -(1-\alpha)\alpha\tfrac{x^\alpha}{y^\alpha}\left[u^2\cdot\tfrac{y}{x^2} - 2uv\cdot\tfrac{1}{x} + \tfrac{v^2}{y}\right] =$$

$$= -(1-\alpha)\alpha\tfrac{x^\alpha}{y^\alpha}\cdot y\left[u^2\cdot\tfrac{1}{x^2} - 2uv\tfrac{1}{xy} + \tfrac{v^2}{y^2}\right] =$$

$$= -\alpha(1-\alpha)\tfrac{x^\alpha}{y^\alpha}y\left[\tfrac{u}{x} - \tfrac{v}{y}\right]^2 \leq 0$$

Thus H is negative semi-definite.
This means that f is concave and quasiconcave.

$S_b = \{(x,y) \mid x^\alpha\cdot y^{1-\alpha} \leq b\ ,\ x,y \in \text{dom } f\}$

$x^\alpha \leq y^{\alpha-1}\cdot b$

- $\alpha = 0 \longrightarrow 1 \leq \frac{b}{y}$, only true for $b \geq y$.

- $\alpha = 1 \longrightarrow x \leq b$, which is a halfspace.

- $0 < \alpha < 1 \longrightarrow x \leq \sqrt[\alpha]{b\cdot y^{-1}}\cdot y \quad , \quad b > 0$

$-x \geq -\sqrt[\alpha]{b\cdot y^{-1}}\cdot y$ Let $u = -x$
$u \geq -\sqrt[\alpha]{b\cdot y^{-1}}\cdot y \quad , u < 0$ This is the epigraph of $h(y) = -\sqrt[\alpha]{b\cdot y^{-1}}\cdot y$

with $h(y) = -c \cdot y^{1-\frac{1}{\alpha}}$, $\alpha \in (0,1)$

Curvature of h:
Since $1 - \frac{1}{\alpha} < 0$, the function $y^{1-\frac{1}{\alpha}}$ is convex.
Thus, $h(y) = -(\text{convex function})$ is concave.

Conclusion: S_b is the epigraph of a concave function. The epigraph of a concave function is not always convex. Therefore, f is not quasiconvex.

3.22

(a) Let $v = \sum_{i=1}^{n} e^{y_i}$. Since exponentials are strictly positive, $v > 0$.
For the outer logarithm to be defined, we require $-\log(v) > 0 \implies \log(v) < 0 \implies v < 1$.
Thus, our required starting input is restricted to $v \in (0, 1)$.

$h_1(v) = \log v \quad \rightarrow \quad \begin{array}{|c|}\hline (0,1) \\ \hline \nearrow \\ \hline \end{array}$ Restricted dom: $(0, 1)$
Range: $(-\infty, 0)$

$\longrightarrow \quad h_2(v) = -\log v = -h_1(v)$.
The range is multiplied by -1 .

$h_2(v) = -\log v \quad \rightarrow \quad \begin{array}{|c|}\hline (0,1) \\ \hline \searrow \\ \hline \end{array}$ dom: $(0, 1)$
Range: $(0, +\infty)$

$\longrightarrow$ Next is the outer function: $f(v) = -\log(h_2(v))$.
Since $h_2(v)$ outputs exactly into $(0, +\infty)$, it matches
the required domain of the outer logarithm.

f(v) = -log(h_2) $\quad \rightarrow \quad \begin{array}{|c|}\hline h_2 \in (0,+\infty) \\ \hline \searrow \\ \hline \end{array}$ Range: $(-\infty, +\infty)$

$-\log(-\log(\sum e^{y_i}))$

$g = \log(\sum e^{y_i})$ is **convex**.

$h_1(x) = -x$ non-increasing, linear.

$h_2 = h_1(g) = -\log(\sum e^{y_i})$ is **concave**.
(Linear non-increasing function of a convex function is concave.)

$h_3 = \log h_2 : \quad \begin{cases} \log \text{ is concave, non-decreasing} \\ h_2 \text{ is concave} \end{cases}$
Thus h_3 is **concave**.

$h_4 = -h_3 : \quad \begin{cases} h_3 \text{ is concave} \\ -x \text{ is non-increasing, linear} \end{cases}$
Thus h_4 is **convex**.

(b) $-\sqrt{u\left(v - \frac{x^T x}{u}\right)}$, $h_1 = \frac{x^T x}{u}$, h_1 is convex

$-\sqrt{u(v - h_1)}$, $h_2 = -h_1$ $\begin{cases} h_1 \text{ convex} \\ -x \text{ non-increasing, linear} \end{cases}$, thus h_2 is concave

$-\sqrt{u(v + h_2)}$, $h_3 = v + h_2$ {non-negative weighted sum preserves concavity} thus h_3 concave

$-\sqrt{u \cdot h_3}$, given that $-\sqrt{x_1 x_2}$ is convex $\begin{cases} \sqrt{\cdot} \text{ is non-decreasing} \\ -x \text{ is decreasing} \\ -\sqrt{\cdot} \text{ is decreasing} \end{cases}$

$\left[-\sqrt{h_3 \cdot u}\right]$, $\begin{cases} u \text{ linear (concave)} \\ h_3 \text{ concave} \end{cases}$, $-\sqrt{x_1 x_2}$ is non-increasing and convex.

From vector composition rules f is convex.

(c) $-\log(uv - x^T x)$

$$-\log((v - \tfrac{x^T x}{u})u) = \underbrace{-\log u}_{h_1} \underbrace{-\log(v - \frac{x^T x}{u})}_{h_2} \qquad \tfrac{x^T x}{u} : \text{convex}$$

Analysis of Component h_1:
$\log x$: concave
$-x$: linear, non-increasing
h_1 : convex

Analysis of Component h_3:
$\frac{x^T x}{u}$: convex
$-x$: linear, non-increasing
$-\frac{x^T x}{u}$: concave (h_3)

Analysis of $v + h_3$:
h_3 concave
v : linear and non-decreasing
$v + h_3$: concave
(preservation on non negative sum)

Analysis of Component h_2:
$v - \frac{x^T x}{u}$: concave
$\log(x)$: concave, non-decreasing
h_2 : concave

Conversion of h_2:
h_2 : concave
$-x$: linear, non-increasing
$-h_2$: convex

Final Combination:
h_1 : convex
h_2 : convex
$h_1 + h_2$: convex
(preservation on non negative sum)

Thus f is convex

(d) $-\left(t^p - \|x\|_p^p\right)^{1/p} =$

$$= -\left(t^{p-1}\left(t - \underbrace{\frac{\|x\|_p^p}{t^{p-1}}}_{g(x,t)}\right)\right)^{1/p}$$

$$= -(t)^{1-\frac{1}{p}} \cdot \underbrace{(t - g(x,t))}_{h}{}^{1/p} = -(t)^{1-\frac{1}{p}} \cdot (h)^{1/p}$$

Definition of g:
$g(x,t) = \frac{\|x\|_p^p}{t^{p-1}}$
Because norm is always non-negative we can use the property for $g(x,t)$ in dom f.

Curvature Analysis (Step 1):
g : convex
$-x$: linear, non-increasing
$-g$: convex

Curvature Analysis (Step 2):
t : linear
$-g$: convex
$t - g$: convex
h : convex

Final Composition Logic:
It is given that: $-(x)^{1/p}(y)^{1-1/p}$ is convex.
Also non decreasing in each component.
h : convex
t : linear (convex)

Thus f is convex.

(e) $-\log\left(t^{p-1}\left(t-\frac{\|x\|_p^p}{t^{p-1}}\right)\right)$

$$= -\log t^{p-1} - \log\left(t-\frac{\|x\|_p^p}{t^{p-1}}\right)$$

$$= -(p-1)\log t - \log\left(t - \underbrace{\frac{\|x\|_p^p}{t^{p-1}}}_{h_1}\right)$$

Curvature Analysis (Step 1):
h_1 : convex
$-x$: linear, non-increasing
$-h_1$: concave

Curvature Analysis (Step 2):
t : linear (concave)
$-h_1$: concave
$t-h_1$: concave

Composition Step 1:
$t-h_1$: concave
$\log x$: concave, non-decreasing
$\log(t-h_1)$: concave

Composition Step 2:
$\log(t-h_1)$: concave
$-x$: linear, non-increasing
$-\log(t-h_1)$: convex

Final Conclusion:
$-\log t$: convex $\quad$ ($\log x$ concave)

$-\log(t - h_1)$: convex
Thus $-\log t - \log(t - h_1)$ **Convex.**

Extra 3.19

$f, g : \mathbb{R}^n \to \mathbb{R}$
$\phi : \mathbb{R} \to \mathbb{R}$

(a) f, g convex, then $h(x, y) = (f(x) + g(y))^2$ convex. FALSE

Let $u(x, y) = f(x) + g(y), \quad u : \mathbb{R}^n \to \mathbb{R}$
$u(x, y)$ is convex as a non-negative sum of convex functions.

Let $v(x) = x^2, \quad \mathbb{R} \to \mathbb{R}_+$

We have $h(x, y) = v(u(x, y))$

v is convex and u is convex, but v is not non-decreasing.

(It would be convex if $f + g : \mathbb{R}^n \to \mathbb{R}_+$)

Counter-example : For $f(x) = x^2$ and $g(y) = -1$: $v(f(x) + g(y)) = (x^2 - 1)^2$
2nd order condition:
$\nabla^2 \left((x^2 - 1)^2\right) \geq 0 \implies$

$\frac{\partial}{\partial x} \left(2(x^2 - 1) \cdot 2x\right) =$

$4(x^2 - 1) + 4x \cdot 2x =$

$12x^2 - 4 \geq 0 \implies x^2 \geq \frac{1}{3}$

False for $x \in \left(-1/\sqrt{3}, 1/\sqrt{3}\right)$

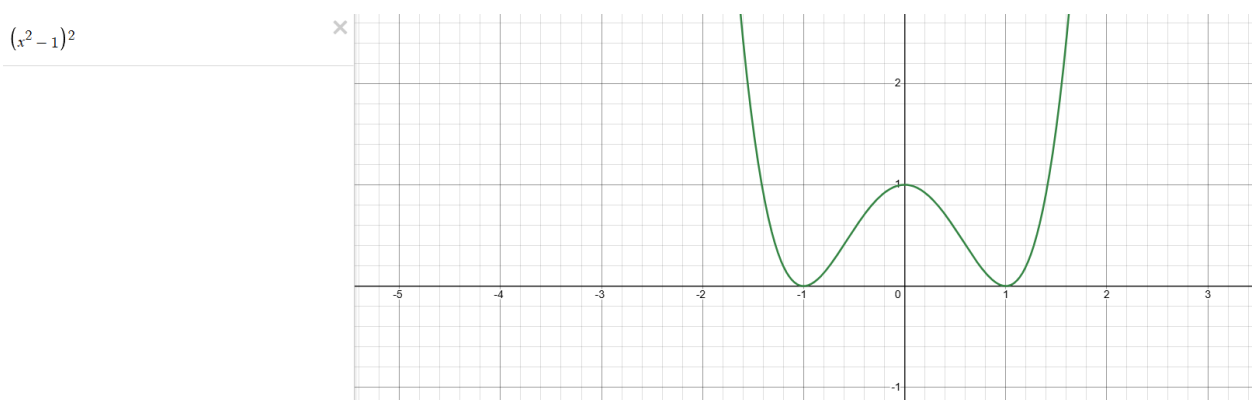

Figure 2: Graphical representation of $f(x) = (x^2 - 1)^2$

(b) $f : \mathbb{R}^n \to \mathbb{R}$ convex, differentiable
$\phi : \mathbb{R} \to \mathbb{R}$ convex, differentiable
$\phi' > 0$

Statement: $\phi(f(x))$ convex. TRUE

Check composition rule of differentiable convex functions:

$g = \phi(f(x))$

ϕ convex and non-decreasing ($\phi' > 0$)
f convex.

Thus g is convex.

(c) From vector composition Rule:
$u(x, y) = \sqrt{xy}, \quad u : {\mathbb{R}_{++}}^2 \to \mathbb{R}$: concave (Geometric Mean)
f, g: concave

$u(x, y)$ non-decreasing :

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(xy)^{1/2} = \frac{1}{2}(xy)^{-1/2} \cdot y = \frac{y}{2\sqrt{xy}} = \frac{1}{2}\sqrt{\frac{y}{x}}$$

Since $x > 0$ and $y > 0$, then $\frac{1}{2}\sqrt{\frac{y}{x}} > 0$. Similarly:

$$\frac{\partial u}{\partial y} = \frac{1}{2}\sqrt{\frac{x}{y}} > 0$$

The function is non-decreasing in each argument.

$h(f(x), g(y)) = \sqrt{f(x) \cdot g(y)}$ is concave.

3.16(b) Proof for quasiconvexity/quasiconcavity

$f(x_1, x_2) = x_1 \cdot x_2 \quad , \quad \mathrm{dom} f = \mathbb{R}^2_{++}$

Sublevel sets: $S_a = \{(x_1, x_2) \mid x_1 \cdot x_2 \leq a \ , \ x_1, x_2 \in \mathrm{dom} f\}$

Let (x_1, x_2) and $(x_1', x_2') \in S_a$

then: $\quad \theta \cdot (x_1, x_2) + (1-\theta)(x_1', x_2') \in S_a \quad \forall (x_1, x_2), (x_1', x_2')$
with $\theta \in [0, 1]$

$(\theta x_1 + (1-\theta)x_1', \theta x_2 + (1-\theta)x_2') \in S_a$

$$\theta x_1 \cdot \theta x_2 + \theta x_1 \cdot (1-\theta)x_2' + (1-\theta)x_2 \cdot \theta x_1' + (1-\theta)^2 x_1' x_2' =$$

$$\theta^2 x_1 x_2 + x_1 x_2' \cdot (\theta - \theta^2) + (\theta - \theta^2) x_2 x_1' + (1-\theta)^2 x_1' x_2' =$$

$$\theta^2 x_1 x_2 - \theta^2 x_1 x_2' - \theta^2 x_2 x_1' + \theta^2 x_1' x_2' + x_1 x_2' \theta + \theta \cdot x_2 x_1' - 2\theta x_1' x_2' + x_1' x_2' =$$

$$\theta^2 (x_1 x_2 - x_1 x_2' - x_2 x_1' + x_1' x_2') + \theta \Big(\overbrace{x_1 x_2' + x_2 x_1' - 2x_1' x_2'}^{x_1 x_2' + x_2 x_1' - x_1' x_2' - x_1' x_2'} \Big) + x_1' x_2' =$$

$$\theta^2 \Big(x_1 (x_2 - x_2') - x_1' (x_2 - x_2') \Big) + \theta \Big(x_1' (x_2 - x_2') + x_2' (x_1 - x_1') \Big) + x_1' x_2' =$$

$$\theta^2 \Big((x_2 - x_2')(x_1 - x_1') \Big) + \theta \Big(x_1' (x_2 - x_2') + x_2' (x_1 - x_1') \Big) + x_1' x_2' \leq a$$

$$\frac{\theta^2 ((x_2 - x_2')(x_1 - x_1'))}{x_1' x_2'} + \theta \left(\left(\frac{x_2}{x_2'} - 1 \right) + \frac{x_1}{x_1'} - 1 \right) + 1 \leq \frac{a}{x_1' x_2'}$$

$$\frac{\theta^2 ((x_2 - x_2')(x_1 - x_1'))}{x_1' x_2'} + \theta \left(\frac{x_2}{x_2'} + \frac{x_1}{x_1'} - 2 \right) + 1 \leq \frac{a}{x_1' x_2'}$$

$\Big[$By choosing points at the boundary : $x_1 x_2 = x_1' x_2' = a \implies \frac{x_1}{x_1'} = \frac{x_2'}{x_2} = 2c \Big]$

$$\frac{\theta^2((x_2 - x_2')(x_1 - x_1'))}{a} + \theta\left(\frac{1}{2c} + 2c - 2\right) \leq 0$$

$$\theta^2 \frac{(x_2 - x_2')(x_1 - x_1')}{x_1' x_2'} + \theta\left(\frac{1}{2c} + 2c - 2\right) \leq 0$$

$$\theta^2 \left(\frac{x_2}{x_2'} - 1\right)\left(\frac{x_1}{x_1'} - 1\right) + \theta\left(\frac{1}{2c} + 2c - 2\right) \leq 0$$

$$\theta^2 \left(\frac{1}{2c} - 1\right)(2c - 1) + \theta\left(\frac{1}{2c} + 2c - 2\right) \leq 0$$

$$\theta^2 \left(1 + 1 - \frac{1}{2c} - 2c\right) + \theta\left(\frac{1}{2c} + 2c - 2\right) \leq 0$$

$$\theta^2 \left(2 - \frac{1}{2c} - 2c\right) - \theta\left(2 - \frac{1}{2c} - 2c\right) \leq 0$$

$$\theta^2 \cdot 2\left(1 - \frac{1}{4c} - c\right) - \theta \cdot 2\left(1 - \frac{1}{4c} - c\right) \leq 0$$

$$q := \left(1 - \frac{1}{4c} - c\right)$$

$$2 \cdot (\theta^2 \cdot q - \theta q) \leq 0$$

$$(\theta^2 - \theta) \cdot q \leq 0$$

$$q = 1 - \frac{1}{4c} - c = \frac{c - \frac{1}{4} - c^2}{c} = -\frac{(c^2 - c + \frac{1}{4})}{c} = -\frac{(c - \frac{1}{2})^2}{c} < 0$$

since $x_1, x_2, x_1', x_2' \in \mathbb{R}_{++}$, this is a negative expression $\left(\frac{x_1}{x_1'} = \frac{x_2'}{x_2} = c > 0\right)$ and distinct points mean $c \neq \frac{1}{2}$.

Also because $\theta \in (0, 1)$, $\theta^2 < \theta \implies \theta^2 - \theta < 0$

Thus $\underbrace{(\theta^2 - \theta)}_{(<0)} \cdot \underbrace{q}_{(<0)} \leq 0$ is a contradiction.

We supposed that the convex combination of any two points in the sublevel set is in the sublevel set S_a (convex set).